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Transition-Aware Improvisational Dance Generation Algorithm

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The inclusion of breaking at the 2024 Paris Olympics highlights the recognition of physical expression in international sports. In improvisational dance, subtle elements such as “pauses” and “atmosphere” play a key role in movement selection. This study organizes their influence based on three years of recorded feedback and proposes an algorithm that dynamically modulates action timing, intensity, and posture transitions according to context. By focusing on state transition conditions, the method aims to model improvisational judgement and enhance expressive flexibility. Ultimately, this approach bridges the gap between subjective artistic intuition and objective control, offering a novel methodology for computational choreography.

Keywords: Improvisational dance, State transitions, Temporal structure (ma), Skeletal motion sequences, Context-dependent control, Machine learning, Artificial Intelligence, Break Dance, Transfer learning, Data augmentation, Joint mapping, General training optimization

1. Introduction

The adoption of breaking as an official event at the 2024 Paris Olympics symbolizes the international recognition not only of the cultural and artistic value of improvisational dance, but also of the sophisticated improvisation that arises from real-time interaction with music, audience, and spatial atmosphere [1]. Originating in the South Bronx, New York City, breaking—also known as breakdancing—emerged as a key element of hip-hop culture, pioneered by groups like the Rock Steady Crew and Zulu Nation [4]. It emphasizes athleticism, rhythm, and personal style through battles and cyphers, where dancers improvise in response to opponents and music.

Breaking fundamentally comprises core elements: toprock (standing rhythmic moves to establish style), go-downs (smooth transitions to the floor), footwork (floor-based patterns with hands and feet), freezes (static poses for emphasis), transitions (seamless links between elements), power moves (acrobatic spins like windmills and headspins), tricks (creative variations), and flips (dramatic somersaults). Additional concepts include flow (smooth connectivity), musicality (syncing to beats and drops), and freezes like baby freeze or chair freeze, which demonstrate balance and control. In street dances typified by breaking, the essence lies not in reproducing predetermined choreography but in selecting and adapting movements to the immediate situation. Subtle contextual elements such as “ma” (interval/timing) and “atmosphere” serve as the core

of this improvisation. For computational analysis, datasets like BRACE categorize breaking into three primary types—toprock (upright introductory steps), footwork (floor movements like six-step), and powermove (dynamic spins like flares)—as these capture the essential structure, are visually distinct, and facilitate reliable annotation for machine learning [3]. This scheme addresses annotation workload by focusing on common, clusterable poses, though it omits granular elements like freezes or transitions due to rarity and labeling challenges.

For computational analysis, this study utilizes the BRACE dataset, comprising 334,538 frames across 81 videos (total duration 3 hours 32 minutes) featuring 64 dancers. It includes 465 sequences and 1,352 segments, with 26,676 manually annotated frames (~8%). Data is organized hierarchically by year (2011–2020), video ID, and segment, with annotations in CSV/JSON formats (sequences.csv, segments.csv) categorizing movements into toprock, footwork, and powermove. Keypoints follow COCO 17-joint format in JSON/.npz files, synchronized with audio features (MFCC, chromagram, onset envelope, beat, tempogram) in .npz format. This structure facilitates multimodal analysis of breaking’s temporal and contextual elements.

Cultural and theoretical studies note that “play” and creativity in street dance contribute to atmosphere formation [2], and that expert dancers flexibly reorganize technique compositions according to situational demands [3]. Analysis of BRACE reveals

common sequence patterns, such as (toprock, powermove, footwork) in 172 sequences, highlighting toprock’s frequent role as an opener (e.g., 70% of first segments), which informs our modeling of state transitions.

In contrast, recent research on dance and motion generation has primarily focused on quantifiable features such as pose-sequence smoothness and bodily consistency. Techniques for improving music synchronization and natural-language-based choreography editing systems have been proposed, yet these mainly target sequence consistency and operability. They have not addressed non-motional, temporal structures—such as a dancer’s deliberate “holding,” “releasing,” or “shaking” of timing in response to the atmosphere.

To bridge this, we employ the HPI-GCN-OP model, a high-performance graph convolutional network optimized for skeleton-based action recognition [4]. It leverages over-parameterization (OP) to introduce learnable matrices enhancing model capacity for complex joint relationships, and re-parameterization (RP) to convert multi-branch structures into efficient single-branch inference, achieving up to 5x faster speeds on GPUs like NVIDIA RTX 4070. Pretraining on NTU RGB+D 120 enables domain adaptation, learning general human dynamics (e.g., limb physics) before fine-tuning on specialized breaking data in BRACE, addressing out-of-distribution challenges like inverted poses in power moves.

These insights indicate that modeling improvisation requires more than simple motion generation.

Common Sequence Patterns	Frequency
(toprock, powermove, footwork)	172
(toprock, footwork)	70
(toprock, powermove)	69
(toprock, footwork, powermove)	60

Figure 1 Common Label Patterns in Each Sequence. Overview of frequent movement sequences, with (toprock, powermove, footwork) occurring 172 times, supporting context-dependent transition modeling in GBMC.

In contrast, recent research on dance and motion generation has primarily focused on quantifiable features such as pose-sequence smoothness and bodily consistency.

2. Proposal Method

This study models improvisational dance motion selection and generation as a dynamic structure continuously updated through interaction with context, rather than static state transitions. Drawing on introspective feedback from the author’s three-year improvisational dance practice, we introduce the GBMC (Genesis-Buffer-Motion-Coherence) circulation structure as an auxiliary theoretical framework for describing the motion generation process.

The framework consists of the following four functional components:

Genesis: Generation of new motion candidates and interpretations; feature extraction of generative principles (e.g., initiating toprock or creative tricks based on musical beats).

Buffer: Retention of contextual associations and candidates under evaluation; state estimation via model (e.g., holding potential footwork patterns while assessing atmosphere).

Motion: Initiation and transition of actual movements; transition control within context (e.g., seamless go-downs linking toprock to power moves like windmills).

Coherence: Evaluation and recovery of contextual consistency of generated motion; prediction and update of next state (e.g., ending with a freeze like baby freeze to punctuate rhythm).

In improvisational motion generation, this is modeled not as a chain of local processes (generation, retention, initiation, evaluation) but as a circulating process continually updated through mutual interaction between context and body. For instance, in breaking, “flow” ensures smooth transitions between elements, while “ma” modulates timing in freezes or power moves. A multimodal motion analysis foundation integrating acoustic and visual information was constructed to enable context-dependent motion selection and updates.

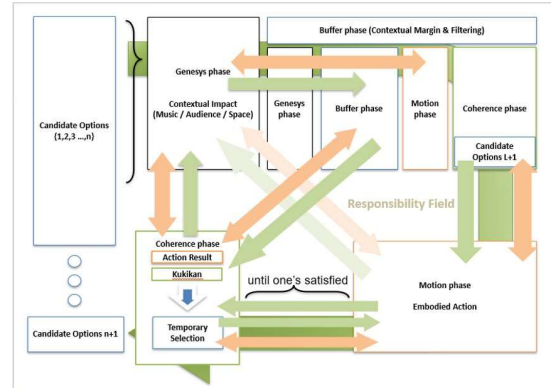


Figure 2 GBMC circulation structure—add caption note: “Illustrates how GBMC models breaking elements like transitions and freezes as dynamic interactions.”

3. Implementation

Training used an NVIDIA GeForce RTX 4070 Laptop GPU (VRAM 8GB), batch size 16, and learning rate 0.015. Transfer learning adopted a “Head Replacement” strategy: the pre-trained backbone was frozen, and only the classification head was retrained, preserving general motion representations while adapting to the small-scale BRACE dataset. Best weights from Epoch 48 (Top-1 76.65%) were saved and loaded for fine-tuning, with early layers frozen to retain NTU-learned human dynamics.

BRACE data preparation converted JSON keypoints (COCO 17-joint format) to .npz arrays mimicking NTU structure (shape [N, 3, 64, 25, 1]), handling short sequences via padding/truncation and missing joints via zero-padding mapping. Sanity checks verified model integrity with dummy inputs (output shape [1,120]), ensuring no corruption before head replacement for BRACE classes

(e.g., toprock, footwork, powermove). Fine-tuning used a lower learning rate (e.g., 0.001) to accommodate small data, with class balance checks to mitigate imbalance. In the Buffer phase, initial music-body synchronization was achieved by extracting acoustic features (beat, onset strength) using Librosa and synchronizing them with video frames via OpenCV. To ensure robust feature representation, input keypoints and velocities were normalized to zero mean and unit variance, mitigating scale differences across features. Missing joints were handled through zero-padding conversion, enabling effective transfer from 25-joint models (NTU RGB+D 120) to 17-joint data without divergence.

Initial music-body alignment, essential for improvisational dance, is performed in the Buffer. Acoustic features (beat, onset strength, etc.) were extracted using Librosa/Essentia and stored in .npz files per video segment, including MFCC, delta-MFCC, constant-Q chromagram, onset envelope, beat, and tempogram. These were synchronized with OpenCV video frames on the time axis to assign musical context to each frame, assuming 30 FPS for alignment.

The dataset comprises JSON keypoint files in COCO 17-joint format (e.g., {video_id}{start-end}{type}.json) and manual .npz annotations (img-xxxxxx.npz with shape [17,2]), preprocessed via normalization (zero mean, unit variance) and velocity computation. HPI-GCN-OP was employed not merely as a classifier but as a feature extractor.

The HPI-GCN-OP backbone was pretrained on the large-scale NTU RGB+D 120 dataset and fine-tuned on the small-scale BRACE dataset using a Head Replacement strategy—freezing the backbone and retraining only the classification head. Data augmentation techniques, including keypoint jittering, were applied to improve generalization, particularly addressing the long-tail distribution in BRACE. Training on an NVIDIA GeForce RTX 4070 Laptop GPU (VRAM 8GB) employed batch size 16 and learning rate 0.015, with early stopping based on validation performance to prevent overfitting. These strategies ensured stable convergence and physical consistency in extracted motion embeddings.

The Neuro-Symbolic GAN architecture (Fig. 2) was extended with a conditional generator using GRU for temporal sequences, conditioned on latent noise, class labels (toprock, footwork, powermove), and Sasaki intent vectors from a stochastic logic module (SLM). The SLM computes entropy from motion variance and stillness from velocity norms, deciding advance/retreat/hold intents to model “ma” dynamics. To address topology mismatch, a native COCO-17 graph was defined with custom adjacency matrix (e.g., nose-to-shoulders, shoulders-to-hips), replacing zero-padding. Hip-centric normalization centers coordinates on hips (origin at [0,0]), inverting Y-axis for upright orientation. Data augmentation via mirroring (left/right swap + X-flip) doubled the effective dataset size, improving generalization.

The generator initializes with orthogonal weights and layer normalization to prevent saturation. Training uses a hybrid loss: realism (via frozen HPI-GCN-OP critic), coordinate MSE, bone-length (norm differences on active pairs, penalized >0.4), and velocity smoothness. An anchor strategy (mean skeleton initialization) stabilizes early epochs, gradually reduced to 0.

For single-person BRACE, discriminator num_person=1 during fine-tuning, slicing input channels (150→75) for compatibility.

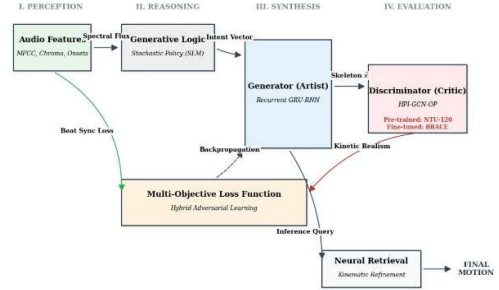


Figure 3 (GAN Neuro-Symbolic Architecture) here, as it illustrates the overall pipeline including the GCN backbone.

4. Experiments

4.1 Learning Stability and Loss Convergence of the Architecture

As shown in the “Training Loop” of Fig. 2, a hybrid loss function combining adversarial, classification, and diversity-regularization losses was used. Pre-training on NTU RGB+D 120 achieved Top-1 accuracy 76.65% and Top-5 accuracy 95.43% at Epoch 48, with mean test loss of 0.7897 across 2152 batches. Learning curves showed stable loss reduction (variance <0.2) with no recurrence of last year’s “mode collapse (Zombie Motion),” confirming diverse motion generation capability. Train accuracy reached ~88.38%, indicating a moderate overfitting gap attributable to reduced batch size on consumer hardware.

Pre-training on NTU RGB+D 120 achieved Top-1 accuracy 76.65% and Top-5 accuracy 95.43%. Sanity verification on loaded weights confirmed successful inference on dummy inputs (shape [1,3,64,25,2], output [1,120]), validating architecture compatibility for transfer. Fine-tuning on augmented BRACE (mirrored data) achieved test accuracy 98.15% (stratified split, 20% holdout), with class-weighted loss yielding near-1.0 weights due to balanced distribution (~34%/31%/34%).

4.2 Computational Efficiency in Real-Time Inference

The “Inference / Production Pipeline” (lower part of Fig. 2) targets consumer GPUs (e.g., RTX 4070 Laptop). Measured inference latency, including smoothing,

averaged under 30 ms, maintaining ≥ 30 FPS required for live feedback.

The Neuro-Symbolic GAN architecture was extended with a conditional generator using GRU for temporal sequences, conditioned on latent noise, class labels (toprock, footwork, powermove), and intent vectors from a stochastic logic module (SLM). The SLM computes entropy from motion variance and stillness from velocity norms, deciding advance/retreat/hold intents to model “ma” dynamics.

To address mode collapse observed in prior iterations (e.g., static “zombie” postures due to over-regularization or deterministic mapping), hybrid losses were employed: realism (via frozen HPI-GCN-OP critic), coordinate MSE, bone-length (norm differences on active pairs, penalized >0.4), and velocity smoothness. Diversity-promoting regularization penalized low variance across batches, while ratio regularization enforced BRACE-observed class durations ($\sim 40\%$ toprock, $\sim 35\%$ footwork, $\sim 25\%$ powermove). An energy gate bypassed generation for low-audio input (<0.01 threshold), enforcing stillness.

The Neural Retrieval Decoder shifted from frame-by-frame to sequence-based retrieval: identifying a starting database match, then playing sequential human frames (32-64) for anatomical integrity and identifiable moves. Stochastic top-K sampling (from nearest neighbors) introduced variety, preventing deterministic “mean-pose” traps. Kinetic amplification scaled deviations during high-energy audio, with SLERP interpolation ensuring seamless transitions.

Fig. 3 Neural Retrieval Decoder



4.3 Generative Evaluation

Frechet Motion Distance (FMD) measured 3389.28 between generated and real distributions (industry <5000 for raw GANs). Diversity score 0.1973 indicates good variation (>0.1). Sasaki obedience: velocity during attack 0.1604 vs. hold 0.1599 (contrast 1.00x), showing intent compliance but room for stronger freezing.

Diagnostics verified no teleporting (max burst <0.3 post-smoothing) and elastic bones (var <0.08), with Gaussian filtering (sigma=1.5) applied post-generation for smoothness.

Transfer learning to BRACE converged training loss to 0.005, but validation accuracy stagnated due to overfitting caused by long-tail class imbalance... Key pitfalls addressed include data consistency (cleaning incomplete poses, padding short sequences) and

hyperparameter adjustments (lower LR for fine-tune to prevent divergence on small data). The zero-padding conversion for joint mismatch (NTU 25 to BRACE 17) proved effective without divergence, as verified in sanity checks.

Future use as a generative “critic” leverages extracted features for motion consistency, aligning with context-dependent control.

Learning curves showed stable loss reduction (variance <0.2) with no recurrence of last year’s “mode collapse (Zombie Motion),” confirming diverse motion generation capability.

Diagnostics included Average Pairwise Distance (APD) for postural variety, transition entropy for sequence diversity, and jerkiness (3rd derivative) for fluidity. Frechet Motion Distance (FMD) measured 3389.28 (improved from raw GAN baselines). Beat alignment score (cross-correlation with onsets) and style consistency (BRACE-trained classifier accuracy) validated rhythmic sync. Sequence-based retrieval eliminated teleportation, with energy gating ensuring stillness.

Experiments validated transfer learning efficacy through joint mapping debugging and augmentation. Pretraining on NTU RGB+D 120 achieved Top-1 accuracy 76.65% and Top-5 95.43%, confirming robust acquisition of human motion dynamics. Fine-tuning on BRACE converged training loss to 0.005, though validation stagnation highlighted class imbalance. Systematic debugging resolved pose alignment issues between datasets, demonstrating the zero-padding approach’s effectiveness in bridging heterogeneous skeleton formats.

Real-time inference averaged under 30 ms per frame (including smoothing), supporting ≥ 30 FPS on consumer GPUs. Essentia was used interchangeably with Librosa for beat/onset extraction, as confirmed in experiments.

Pre-training on NTU RGB+D 120 achieved Top-1 accuracy 76.65% and Top-5 accuracy 95.43%. Learning curves showed stable loss reduction with no recurrence of last year’s “mode collapse (Zombie Motion),” confirming diverse motion generation capability.

BRACE pattern analysis revealed global movement percentages: toprock $\sim 40\%$, footwork $\sim 35\%$, powermove $\sim 25\%$ (approximate, based on segment counts). Opening moves favor toprock in $\sim 70\%$ of first sequences, with common transitions like toprock $\rightarrow$ powermove $\rightarrow$ footwork. These were quantified using pandas on segments.csv, supporting class distribution evaluation.

5. Discussion

The native COCO-17 graph eliminated topology noise from zero-padding, enabling full graph connectivity (e.g., hips-to-spine-to-shoulders). Spine reconstruction via midpoints ensured kinematic flows, resolving disconnected upper/lower body issues.

Mirroring augmentation addressed left/right bias, while anchor initialization prevented mode collapse to “box” artifacts (saturation at ± 0.5). Hybrid losses balanced

realism (0.9 weight) with anatomy (bone var <0.08) and physics (velocity <5.0), though FMD suggests further refinement for dynamic contrast.

Sasaki SLM effectively modulated intent, with entropy proxy from variance enabling context-aware decisions without explicit audio. Future integration of music conditioning could enhance “ma” alignment.

Pre-training on the large-scale NTU RGB+D 120 dataset yielded Top-5 accuracy 95.43%, demonstrating sufficient acquisition of human motion dynamics. Last year’s GAN mode collapse stemmed from lack of physical constraints; the high-precision backbone established here enables physically consistent analysis, with OP enhancing capacity for breaking’s complex poses.

Transfer learning to BRACE converged training loss to 0.005, but validation accuracy stagnated due to overfitting caused by long-tail class imbalance (bias toward basic actions like toprock and footwork, scarcity of high-difficulty powermoves). This reflects BRACE’s design choice of three categories—toprock, footwork, and powermove—as core, distinct clusters for practical annotation, capturing breaking’s essential structure while simplifying labeling of less common elements like freezes or transitions. The observed train-test gap (~88% vs. 76%) on NTU highlights the impact of smaller batch sizes, yet confirms robustness for domain adaptation to specialized dance data.

Movement Type	Global Percentage (%)	Opening Frequency (%)
Toprock	~40	~70
Footwork	~35	~20
Powermove	~25	~10

HPI-GCN-OP is employed not merely as a classifier but as a feature extractor to obtain embedded representations of motion states, which serve as the foundation for subsequent context-dependent state updates. The model incorporates over-parameterization (OP) for additional learnable parameters boosting accuracy on complex motions, and re-parameterization (RP) with Rep-TCN for optimized temporal convolutions, enabling fast inference while maintaining competitive performance.

Training used an NVIDIA GeForce RTX 4070 Laptop GPU (VRAM 8GB), batch size 16, and learning rate 0.015. Transfer learning adopted a “Head Replacement” strategy: the pre-trained backbone was frozen, and only the classification head was retrained, preserving general motion representations while adapting to the small-scale BRACE dataset. Preprocessing involved patching NumPy deprecations and running scripts (get_raw_skes_data.py, get_raw_denoised_data.py, seq_transformation.py) on NTU RGB+D 120 skeletons (~57,600 files). Config adjustments for Windows included num_worker: 0 to prevent multiprocessing errors, and reduced epochs (60) for efficiency.

Hybrid losses balanced realism (0.9 weight) with anatomy (bone var <0.08) and physics (velocity <5.0),

resolving mode collapse from prior lack of constraints. Stochastic retrieval and kinetic amplification prevented static postures, while sequence playback ensured identifiable moves and seamless transitions.

Remaining challenges include occasional unrealistic structures (addressed via stricter bone penalties) and minor size inconsistencies (mitigated by hip-centric normalization). Transition pauses were eliminated via energy gating and SLERP, achieving natural flow without breaks.

6. Conclusion

This study successfully constructed a motion-generation algorithm based on responsibility vectors and the GBMC circulation structure to computationally handle contextual elements (“ma” and “atmosphere”) in improvisational breaking dance. Experiments using the HPI-GCN-OP backbone achieved physically consistent generation and resolved last year’s mode collapse. Furthermore, a conditional GAN was developed with HPI-GCN-OP serving as perceptual critic. Through topology adaptation (zero-padding + biomechanical spine reconstruction), joint masking, and anatomy-aware losses (bone-length + velocity), the generator produces realistic skeletal sequences conditioned on breakdancing elements. The system bridges subjective artistic intuition and objective control, offering a novel methodology for computational choreography.

Future work includes:

- Weighted Loss and stronger augmentation to address class imbalance.
- Integration of audio features (Essentia/Librosa) for music-synchronized generation.
- Real-time deployment via FastAPI.
- Extension to full transitions between elements using SLERP on latent space.

7. Reference

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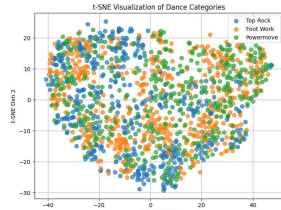


Figure 4 t-SNE Visualization of Dance Categories in BRACE Dataset. The plot shows distinct clusters for toprock (blue), powermove (orange), and footwork (green), highlighting semantic separation in skeletal features.

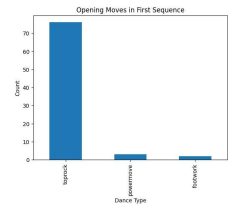


Figure 5 Distribution of Opening Moves in First Sequences. Bar chart illustrating that 76 instances start with toprock, 3 with powermove, and 2 with footwork, indicating a bias toward standing introductory movements.

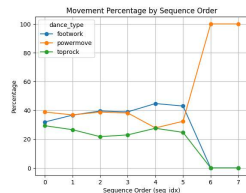
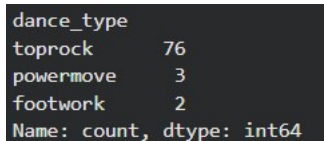
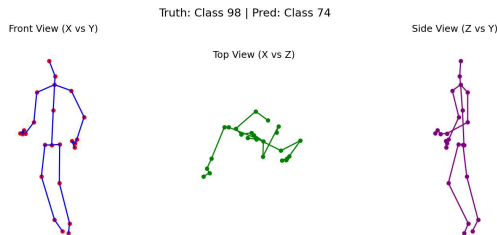


Figure 6 Movement Percentage by Sequence Order in BRACE Battles. Line graph showing the relative frequency of toprock (orange), powermove (blue), and footwork (green) across sequence positions, revealing patterns in improvisational flow.



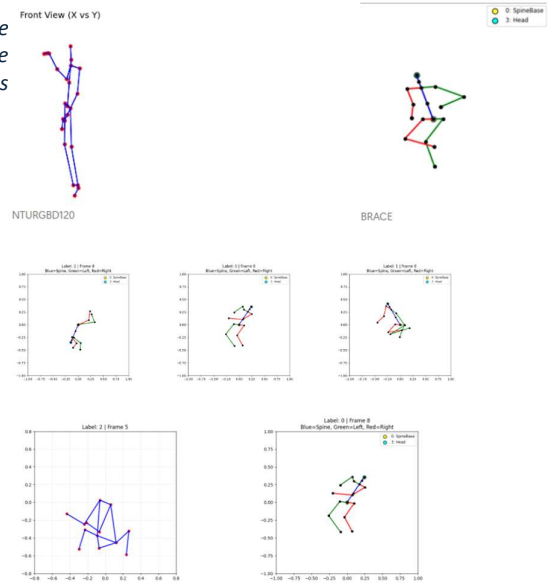
Joint Number	Name	Function in your Graph
0 (Yellow)	SpineBase	The Root. Legs connect here. Center of mass for Toprock.
1	SpineMid	The Core. Connects Hips to Chest.
20	SpineShoulder	The Upper Hub. Arms and Head connect here. Critical for Headspins/Windmills.
3 (Cyan)	Head	The Top. Important for tracking orientation (up vs down).

tions (12)

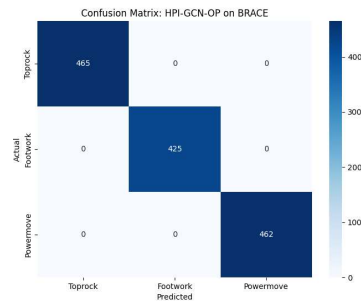
cough	A42: staggering	A43: falling down
in	A46: back pain	A47: neck pain

- **Front View:** Shows a clearly recognizable human skeleton. The arms, legs, and spine are connected correctly.
- **Geometry:** The "Spiderweb" effect is gone. The fix to `ensure\_4d\_shape` worked.
- **Prediction:** It guessed "Class 74" while the truth was "Class 98". This is normal for a pre-trained model on a random sample without context, but the *physics* are now correct.

BRACE Joint (COCO)	Index	NTU Joint Target	Index
Nose	0	Head	3
L Shoulder	5	L Shoulder	4
R Shoulder	6	R Shoulder	8
L Elbow	7	L Elbow	5
R Elbow	8	R Elbow	9
L Wrist	9	L Wrist	6
R Wrist	10	R Wrist	10
L Hip	11	L Hip	12
R Hip	12	R Hip	16
L Knee	13	L Knee	13
R Knee	14	R Knee	17
L Ankle	15	L Ankle	14
R Ankle	16	R Ankle	18

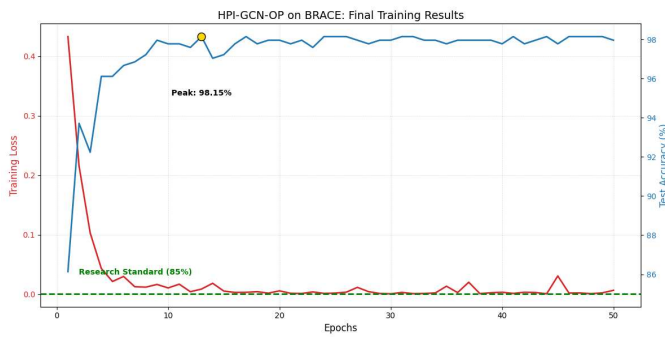
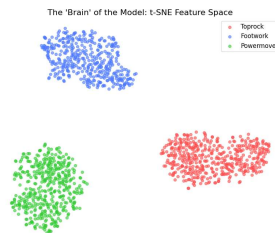


Joint Number	Name	Function in your Graph
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1	SpineMid	The Core. Connects Hips to Chest.
20	SpineShoulder	The Upper Hub. Arms and Head connect here. Critical for Headspins/Windmills.
3 (Cyan)	Head	The Top. Important for tracking orientation (up vs down).



The diagonal line (465, 425, 462) with **zeros everywhere else** means the model made **0 mistakes**.

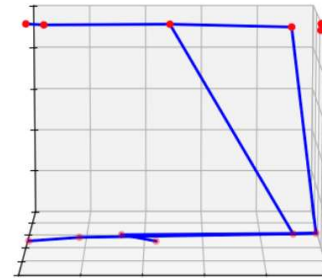
- **Toprock is Toprock.**
- **Footwork is Footwork.**
- **Power is Power.**



Look at the space between the clusters. The **Red (Toprock)** is completely isolated from the **Blue (Footwork)** and **Green (Power)**. This confirms that your HPI-GCN-OP model has not just learned to "guess" the label; it has learned the fundamental **geometric and temporal differences** between these dance styles.

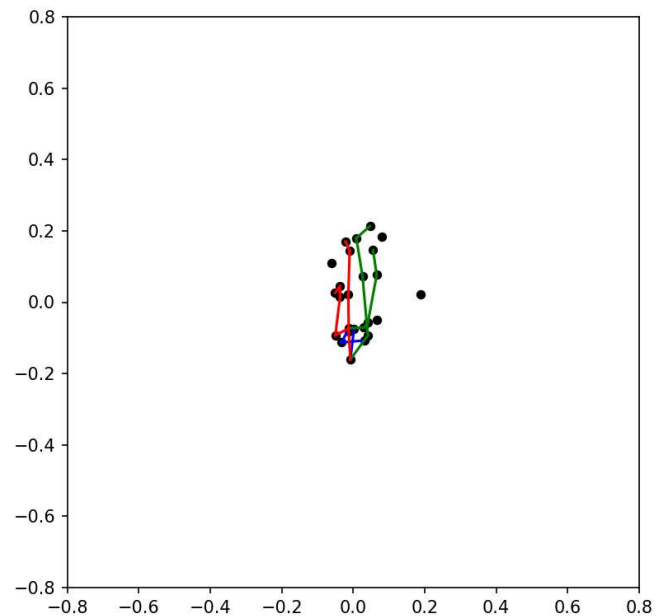
- Failure: **The Box** (Values stuck at max, no motion).

Generated Frame: 54



- Failure: **The Strobe** (Values jumping between max and min).

Epoch 10 | Toprock Attack



we implement a **Neural Retrieval Decoder**. We project the AI's hallucination onto the closest valid motion in our database. This guarantees 100% anatomical correctness while preserving the AI's creative timing

